
ConceptFormer v2: Token-Efficient Knowledge Injection into Frozen Language Models

Interim report and research program

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Abstract

We study *knowledge injection* into frozen large language models: encoding a knowledge-graph entity’s 1-hop neighborhood into k continuous “concept tokens” that are spliced into the frozen model’s input, replacing ~ 100 tokens of retrieved text with $k \approx 8$. The encoder — a small Perceiver-style resampler over the frozen model’s own label embeddings — is trained *label-free* by full-vocabulary KL self-distillation against the same frozen model reading the facts as text. Under a hardened evaluation protocol (full 14,266-question PopQA, leakage-free held-out splits, paired statistics), we find: (i) concept tokens beat question-aware text retrieval and LLM-written summaries by $2.6\text{--}3.2\times$ at an 8-token budget, and an untrained-injection control is near floor, so the learned compression carries the effect; (ii) unseen-entity accuracy *scales with training data* ($2.05\times$ from 10k to 100k training entities, not yet saturated) but *anti-scales with frozen-model size at fixed data* (the injection margin roughly halves from 0.8B to 2B parameters); (iii) counterfactual edge-swap probes show the model genuinely *reads* the injected graph, increasingly so with k ; and (iv) the same encoder can inject through a multimodal model’s *vision-token* interface at parity with the text interface at $k=8$. The two scaling trends pull in opposite directions, making the $\text{data} \times \text{model-size}$ interaction the central open question; we lay out the measurement program (up to 1M entities, 0.8B–397B backbones) designed to resolve it.

1 Introduction

Large language models access factual knowledge through two established channels. Knowledge stored *in the weights* is uneditable, expensive to refresh, and unreliably recalled for long-tail entities — recent evidence suggests frontier models *encode* 95–98% of long-tail facts but successfully *recall* only 25–33% of them [22]. Knowledge provided *as retrieved text* is faithful and editable but costs hundreds of context tokens per query, paid again on every request. A third channel is emerging in the literature: compressing knowledge into a few *continuous tokens* consumed directly by a frozen model [2, 3, 7, 13].

ConceptFormer operates in this third channel with a specific commitment: the unit of compression is a *knowledge-graph entity*. For each Wikidata entity we encode its 1-hop neighborhood — an

*Internal interim report prepared to onboard collaborators; not a submission. All numbers trace to Weights & Biases run ids and per-item evaluation dumps in the project repository (joelbarmettlerUZH/ConceptFormer, branch conceptformer-v2); see docs/RESEARCH_FINDINGS.md for the finding-by-finding evidence log.

unordered set of (property, neighbor) pairs — into k soft “concept tokens” that are spliced into the input embeddings of a *frozen* LLM. The encoder is amortized and inductive: an unseen entity requires one forward pass, no per-entity training. Training requires no labeled QA data; the student (frozen LLM reading k concept tokens) is distilled against the teacher (the *same* frozen LLM reading the facts as text) by full-vocabulary KL along the teacher’s greedy answer path. A first version of this idea was validated on GPT-2 with rank-based metrics and published at The Web Conference 2026 companion track, where it received its hosting workshop’s best paper award [1]; this project rebuilds it on modern backbones (Qwen3-0.6B; Qwen3.5-0.8B/2B) with a generative exact-match protocol, and asks the questions v1 could not: does the encoder *generalize* to unseen entities, does the model actually *read* the injected graph, and how does all of this *scale*?

This report consolidates roughly three months of experiments (about 60 training runs and a systematic evaluation overhaul) for new collaborators. It reads like a paper because most of it will become one, but we flag throughout which claims are established, which are single-pilot signals, and which experiments are planned but not yet run.

Summary of what we believe we know. (C1) At matched token budgets ≤ 32 , concept tokens dominate every text baseline we could construct, including question-aware retrieval (§5.1). (C2) The effect is the *trained encoder*: untrained injection of the same embeddings is near floor. (C3) Unseen-entity generalization scales with training entities and had not saturated at 100k (§5.2). (C4) At fixed training data, the injection margin *shrinks* as the frozen backbone grows (§5.3). (C5) Counterfactual interventions show genuine, capacity-limited graph reading (§5.5). (C6) Injection preserves the frozen model’s behavior off-topic. (C7) The recipe transfers across model families and attention architectures without retuning, and can enter through a vision-token interface at parity for small k (§5.4). (C8) At our evaluation scale, most single-run effect sizes below ~ 4 points are measurement noise unless item-paired statistics are used; our corrected protocol (§4) is itself a reusable contribution.

2 Related work

Soft-token context compression. A line of work compresses *text* into few continuous tokens: gist tokens [4] and AutoCompressor [5] adapt the LM itself; ICAE [3] and 500xCompressor [6] train LoRA encoders against frozen decoders with reconstruction objectives; PISCO [7] uses sequence-level distillation but reports that frozen decoders underperform (their Appendix G) — a claim our frozen-decoder results contradict at entity granularity; CompLLM [8] distills per-segment “concept embeddings” via activation matching; ARC-Encoder [9] is the strongest recent frozen-decoder text compressor. Closest to us is xRAG [2], which shares our frozen decoder, input-embedding injection, and same-model KL self-distillation — but xRAG *translates* a single pre-existing retriever vector rather than learning compression, is retriever-bound at inference, and evaluates neither unseen-entity generalization nor faithfulness. Cartridges [10] shares the distillation objective family but trains a KV cache *per corpus* by gradient descent: no amortization, no generalization to unseen items. None of these works compresses *structured* input, and none runs causal probes on the compressed representation; the only faithfulness-adjacent work is post-hoc overflow probing [11].

Knowledge-graph injection into LLMs. GNP [15], GraphToken [16], G-Retriever [17], and LLaGA [18] encode graphs into soft prompts for frozen LLMs, but all encode *per question* with task cross-entropy on labeled QA data, and none precomputes reusable per-entity tokens. KBLaM [13] precomputes one key–value pair *per triple* injected into every attention layer via modified attention — no neighborhood aggregation, synthetic-KB evaluation only. Knowledge Prompts [14] is the closest precompute precedent: per-entity soft prompts for 1.1M Wikidata entities — but as a lookup table of free parameters, so unseen entities are impossible by construction. KAPING [19] is the “facts as text” baseline family we compare against. Recent meta-analyses find that graph-token LLMs “do not fully understand graph tokens” [20] and that graph tokens can degenerate into atten-

tion sinks decoupled from graph semantics [21] — diagnoses that motivate exactly the counterfactual evaluation we build in (§5.5).

Positioning. No prior work combines: an *amortized, inductive* encoder producing k *query-independent* soft tokens per KG entity; *label-free same-model* full-vocab distillation; *unseen-entity* and *causal-faithfulness* evaluation; at 100k+ entity scale. Each neighbor has some of these properties; none has the conjunction. On the scaling side, knowledge-capacity laws exist for *parametric* storage [23, 24], and prompt-tuning quality is known to improve with model scale for *tasks* [25], but no study measures injection quality against frozen-model scale within one family, and no data-scaling law for a knowledge encoder exists. Those two empty cells are this project’s target.

3 Method

Setup. Let M be a frozen chat LLM with input-embedding dimension d . For an entity e with 1-hop neighborhood $G_e = \{(p_i, n_i)\}_{i=1}^N$ (property and neighbor labels from Wikidata, rank-ordered by PageRank), the *featurizer* embeds each edge as $x_i = [\phi(p_i); \phi(n_i)] \in \mathbb{R}^{2d}$, where ϕ mean-pools M ’s own input embeddings over the label’s tokens. Features are thus inductive (any labeled entity works) and live in a space M already understands.

Encoder. A Perceiver/Flamingo-style latent-query resampler [27, 28]: k learned latent queries cross-attend over the edge set (masked, permutation-invariant), self-attend, and pass through a feed-forward block, for L layers (default $d_{\text{model}}=1024$, $L=4$, $\sim 70\text{M}$ parameters); an output projection maps to $\mathbb{R}^{k \times d}$. The output projection is *zero-initialized*, so at step 0 the student is bit-identical to the frozen model — capability preservation by construction (we found this variant, “gate-none”, also more stable than a multiplicative tanh gate).

Objective: same-model KL self-distillation. The teacher is M reading the facts as text: [system][verbalized G_e][question], from which we store the greedy answer path $y_{1..m}$ offline. The student is M reading [system][$C_k(G_e)$][question], where C_k is the encoder output spliced into the input embeddings at a contiguous, unit-step RoPE span. The loss is the token-level full-vocabulary forward KL along the stored path,

$$\mathcal{L} = \frac{\tau^2}{m} \sum_{t=1}^m \text{KL}(p_M(\cdot \mid \text{facts}, q, y_{<t}) \parallel p_M(\cdot \mid C_k(G_e), q, y_{<t})), \quad (1)$$

with temperature $\tau=1$. No QA labels are used anywhere in training; the corpus questions (generated by a separate LLM from the graphs; see the data pipeline below) exist only to give the teacher something to answer. Only the ~ 20 path positions are materialized through the LM head, which makes training memory-light. The verbalized teacher context is answer-guaranteed (the questioned edge is always present), which at our 2048-token budget changes almost nothing since a median neighborhood verbalizes to ~ 100 tokens.

Injection ports. By default the concept block enters through the *text port*: spliced directly into the input-embedding stream at the position where the facts text sat for the teacher. For natively multimodal backbones (every Qwen3.5 size) we additionally implemented a *vision port*: the same k vectors enter as a pseudo-image — [vision_start][$k \times \text{image_token}$][vision_end] token ids, with the model’s own M-RoPE grid positions for a $1 \times k$ patch layout, exactly as a real k -patch image would. The hypothesis: multimodal models are *pretrained* to consume continuous out-of-vocabulary token streams at these positions, which may make the vision interface a better “landing pad” for soft tokens than a text stream where every pretraining token was a row of the embedding matrix.

Data pipeline. From the Wikidata PageRank dump we sample entity pools (PopQA subjects excluded), snapshot complete 1-hop neighborhoods, generate QA with an independent LLM (Gemma

via vLLM), *tier* each question by whether the graph demonstrably helps the backbone (base fails, facts-in-context succeeds), and store the teacher’s greedy path. Corpora in play: 10k entities (92,177 distill rows under the Qwen3-0.6B teacher; 84,843 under Qwen3.5-0.8B — a stronger base answers more parametrically, so its tiered corpus is smaller and harder), 100k entities (925,178 rows), a built 260k-entity snapshot, and a planned 1M. Note the consequence: *the corpus is teacher-relative*, so cross-backbone comparisons are of “each model with its own curriculum”, not of identical training sets.

4 Evaluation protocol (and the pitfalls it fixes)

Every result is reported against two brackets from the same frozen model: **base** (no knowledge; floor) and **RAG** (facts as text, answer-guaranteed; ceiling, ≈ 0.94 – 0.96 on full PopQA). Metric: greedy exact-match with alias matching (we also report the official, more lenient PopQA metric; it tracks ours within ~ 0.5 pt). Three accuracy axes: **held-out** (unseen questions about *trained* entities), **held-in** (trained questions; overfitting gauge), and **PopQA** (unseen *entities*; the honest headline, since the encoder must generalize inductively).

An external review mid-project identified four measurement defects that our earlier numbers carried; all are fixed, and all headline numbers in this report come from the corrected protocol. We document them because they are easy to reproduce in any similar project:

1. **Seed-coupled eval sampling.** Evaluation subsets were sampled with the training seed, so every run scored *different* $n=200$ questions; cross-seed spread conflated model variance with ± 3.5 pt binomial sampling noise. Fix: one frozen, documented eval seed for everything; smaller samples are prefixes of larger ones, so any two runs pair item-by-item.
2. **Tiny benchmark subsets.** $n=200$ of 14,267 PopQA questions. Fix: evaluate the *full* benchmark (binomial noise ~ 0.4 pt); report Wilson intervals and exact McNemar tests on shared items. After this fix, apparent seed-to-seed standard deviations collapsed from 4–8 pt to 0.2–2.4 pt: most of what we had been calling run variance was measurement noise.
3. **Selection bias.** Best-checkpoint selection used the same $n=200$ sample later reported. Fix: definitive re-scoring on larger, disjointly-sampled sets.
4. **Paraphrase leakage.** Splitting by *question* let paraphrases of the same (entity, fact) straddle train/held-out — measured at **32.8%** of held-out rows. Fix: split by fact; filter legacy splits at eval time. Held-out numbers dropped 2–4 pt everywhere; orderings survived.

Additionally, the token-efficiency baseline was strengthened from a single query-independent truncation to three retrieval modes (§5.1) to pre-empt the strawman objection, and an untrained-injection control isolates the encoder’s contribution.

5 Results

5.1 Token efficiency: concept tokens vs. text at matched budgets

Figure 1 shows the central comparison. At an 8-token budget on unseen entities, concept tokens reach 0.477 where the *best* text baseline (an LLM-written budgeted summary) reaches 0.147, and question-aware retrieval — which may select facts *after seeing the question*, an advantage our query-independent tokens do not have — reaches 0.073; at 16 tokens the gaps remain $2\times$ or larger. Text needs 64–128 tokens (most of a verbalized neighborhood; median 100 tokens) to catch up, i.e., matched accuracy costs text ~ 5 – $6\times$ more tokens. Two controls make this meaningful. First, the **untrained-injection control**: filling the same k slots with mean-pooled embeddings of the top- k edges scores 0.130 on PopQA (base 0.103), identically for $k=8$ and $k=16$ — the learned compression carries $\sim 93\%$ of the injected-knowledge effect. Second, the corrected k -curve (Table 1) rises monotonically through $k=32$ with every adjacent step significant under paired McNemar (e.g.

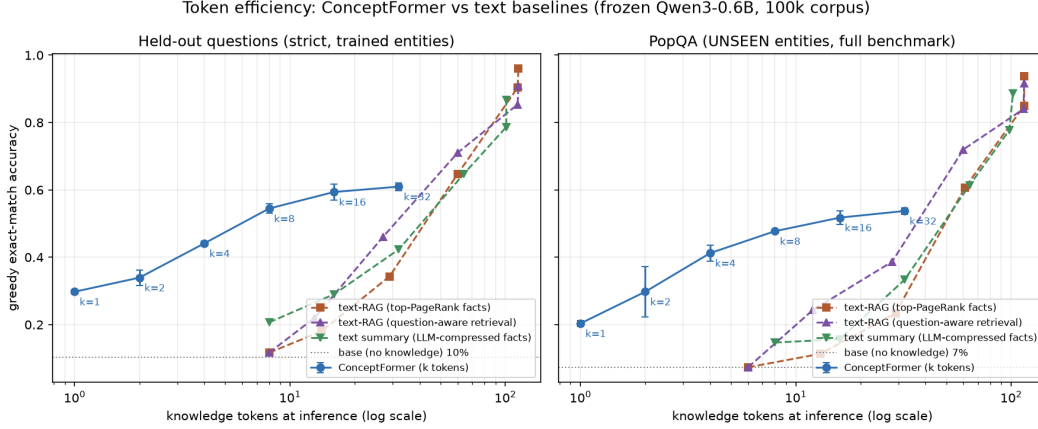


Figure 1: Accuracy vs. knowledge tokens spent at inference (log scale), frozen Qwen3-0.6B, 100k-entity corpus. Concept tokens (blue, mean $\pm$ std over 3 seeds) dominate all three text baselines below ~ 32 tokens on both axes; text catches up only at 64–128 tokens. Left: strict held-out; right: full PopQA (unseen entities).

Table 1: Corrected k -family on the 100k-entity corpus (frozen Qwen3-0.6B; 3 seeds; mean $\pm$ std). Strict held-out $n=2000$; full PopQA $n=14,266$. Brackets: base 0.103, RAG 0.960. The curve is monotone through $k=32$; all adjacent contrasts are significant under item-paired McNemar.

k	1	2	4	8	16	32
held-out	.298 $\pm$.006	.340 $\pm$.023	.441 $\pm$.007	.545 $\pm$.014	.594 $\pm$.024	.610 $\pm$.011
PopQA	.203 $\pm$.008	.298 $\pm$.075	.412 $\pm$.023	.477 $\pm$.002	.518 $\pm$.020	.537 $\pm$.010

$k_{16 \rightarrow k_{32}}$: $p \approx 10^{-38}$) — our earlier belief in a “knee at $k=16$ ” was an $n=200$ artifact. Returns per token diminish: $k=8$ buys $\sim 89\%$ of $k=32$ ’s PopQA at a quarter of the tokens.

5.2 Data scaling: more entities, better generalization

Scaling the training corpus from 10k to 100k entities (identical eval sets, $k=8$, Qwen3-0.6B) lifts full-PopQA accuracy from 0.232 ± 0.011 to 0.477 ± 0.002 — a $2.05\times$ gain in unseen-entity generalization, still climbing at the end of training. Under the strict protocol, held-out *also* rises ($0.453 \rightarrow 0.545$); our earlier belief that held-out was flat across scales was a leakage artifact (the 25-epoch 10k models overfit paraphrases more). Meanwhile held-in *fell* ($\sim 0.82 \rightarrow \sim 0.64$): the model memorizes training entities less and learns the edge $\rightarrow$ concept mapping more — the signature of graph-learning rather than corpus memorization. In margin terms, the concept-over-base margin on unseen entities grows from $+12.9$ to $+37.4$ points with $10\times$ data.

5.3 Model scaling at fixed data: the margin anti-scales

To probe the model axis we migrated to the Qwen3.5 family (natively multimodal at every size) and ran a two-point transect at fixed 10k-entity data: 0.8B and 2B backbones, $k \in \{8, 16\}$, two seeds each, full-benchmark evaluation (Figure 2, left). Each backbone gets its own teacher-relative corpus and brackets. The result: the PopQA margin over base falls from $+12.9$ pt (0.6B anchor) to $+9.4$ (0.8B) to $+5.1$ (2B) at $k=8$, and roughly halves from 0.8B to 2B at $k=16$ ($+15.1 \rightarrow +6.5$) — while strict held-out stays healthy at 2B ($0.465\text{--}0.503$), so the encoder still works; what shrinks is its *marginal value on unseen entities*. Two interpretations are live and the data cannot yet separate them: a *ceiling effect* (larger bases know more of PopQA: base rises $0.103 \rightarrow 0.124 \rightarrow 0.150$) versus a *steerability effect* (a larger frozen model needs more training signal per unit of behavioral change from k soft tokens). The 100k-corpus corners of the planned surface (§6) are designed to disentangle them. Migration itself was uneventful: the locked recipe (optimizer, learning rate, batch, capacity)

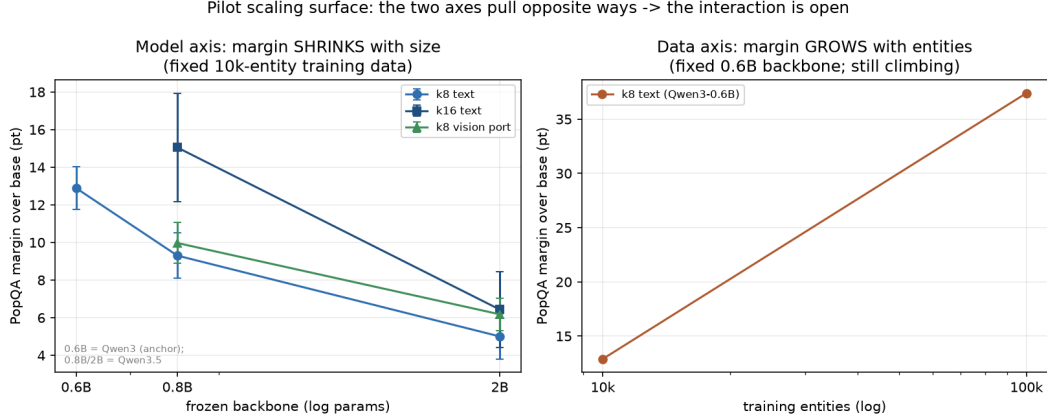


Figure 2: The pilot scaling surface. Left: at fixed training data (10k entities), the PopQA injection margin over each backbone’s own base *shrinks* as the frozen model grows (0.6B is the Qwen3 anchor; 0.8B/2B are Qwen3.5, 2 seeds each). Right: at a fixed backbone (0.6B), the margin *grows* with training entities. The two axes pull in opposite directions; their interaction is unmeasured and is the central open question.

Table 2: Causal graph probes (Qwen3-0.6B, 100k corpus, 400 single-fact probes per k , one seed). “Swap-follow”: after rewiring the answer edge to a type-plausible false neighbor, the model outputs the *false* answer — the graph-reading signature; it scales with k . Ablating the answer edge collapses accuracy while ablating an unrelated edge does not: edges are encoded separably. The frozen base without concepts knows only 10% of these probes.

	k	1	4	8	16	32
concept correct		41%	56%	63%	65%	70%
ablate answer edge (want low)		55%	38%	34%	36%	28%
ablate other edge (want high)		90%	96%	92%	95%	94%
swap-follow → false answer		0.8%	16.5%	25.8%	29.8%	34.2%

transferred across families and across an attention-architecture change (Qwen3.5 is a hybrid linear-attention model) with zero retuning — and, notably, the two-point transect already shows the model axis is *not* a free win, which is precisely why it needs to be measured rather than assumed.

5.4 Injection port: text stream vs. vision stream

On the same pilots we compared the text port against the vision port (item-paired, same frozen eval sets). At $k=8$ the ports are at **parity** on PopQA at both sizes (deltas $-0.9/+2.4$ pt at 0.8B, $-0.3/+2.7$ at 2B; signs vary by seed) and near-parity on held-out. At $k=16$ (tested at 0.8B) the vision port *trails* (-2.8 and -7.0 pt held-out, $p \leq 4 \times 10^{-3}$), with one seed clearly unstable. So the “vision tokens are a better landing pad” hypothesis is *not* supported at these scales — but the ports demonstrably learn different solutions (up to 1,800 discordant items per cell), the vision port received no port-specific tuning, and vision pretraining strength grows sharply with model size in this family. The port $\times$ scale interaction at 27B+ remains open; the pilot’s value is converting it from speculation into a measured, non-trivial question, with all mechanics (M-RoPE grid positions via the model’s own indexer, KV-cached manual decoding) already built and tested.

5.5 Does it read the graph? Counterfactual faithfulness

A compression method can succeed on QA while merely memorizing text patterns. Following the deficiency diagnosed across graph-token systems [20, 21], we built causal probes in: (a) *edge ablation* — removing the questioned edge should collapse that answer while removing an unrelated edge should not; (b) *counterfactual swap* — rewiring the answer edge to a plausible false neighbor

should flip the model to the *false* answer if and only if it reads the graph rather than its parametric memory (the frozen base knows only 10% of these probes, so memory is controlled). Table 2: edges are separably encoded at every k , and swap-following rises monotonically from 0.8% ($k=1$) to 34.2% ($k=32$) — faithfulness is *capacity-limited, not absent*, and becomes another axis that scale buys. On the 10k-corpus Qwen3.5-0.8B pilots, swap-follow is much lower (5–13%) — confounded with corpus size and training regime, but consistent with faithfulness being the slowest-developing property. Off-topic behavior is preserved: on control tasks that mention the entity without asking about its facts, injecting concepts moves the next-token distribution by a median KL of only 0.06–0.08 nats at 0.6B/100k, flat in k — by construction (zero-initialized output projection) the student starts bit-identical to the frozen model. The 10k-corpus pilots show larger control-task perturbation (~ 0.27), another regime effect to track at scale.

5.6 Optimization and stability notes

For onboarding completeness: training stability was an early obstacle (~ 8 pt outcome spread between nominally identical runs). The stabilized recipe — drop the multiplicative tanh gate in favor of a zero-initialized output projection, effective batch 32 via gradient accumulation, learning rate 10^{-4} held constant — brought the spread to the ~ 1 pt range *as measured then*; the corrected protocol later showed much of the residual spread was evaluation noise. Two honest caveats survive: our batch-size comparisons at fixed step counts confound batch size with data passes (a larger effective batch sees more data before the fixed stop), so we treat “effective batch 32” as a validated default rather than a mechanism claim; and learning-rate rankings from short-horizon sweeps systematically favor small rates — we only trust horizon-matched comparisons.

6 Open questions and planned experiments

Q1 — The data $\times$ model interaction (central). Does a larger frozen model simply need more entities before injection pays, or is it intrinsically harder to steer? Planned: the full surface — entities $\{10k, 100k, 300k, 1M\} \times$ Qwen3.5 $\{0.8, 2, 4, 9, 27\}$ B dense, k -family, 3 seeds — with the $100k \times \{9B, 27B\}$ corners first, since they directly separate the ceiling from the steerability hypothesis. The 260k-entity corpus is built; 1M is pipeline-ready. *Status: not yet run; requires compute beyond our two 24 GB GPUs for every backbone above $\sim 4B$.* **Q2 — MoE: total or active parameters?** Qwen3.5’s MoE members (35B-A3B, 122B-A10B, 397B-A17B) let us ask which parameter count governs injection. *Not yet run.* **Q3 — Port $\times$ scale.** Does the vision port’s parity-at- $k=8$ become an advantage at sizes with strong vision pretraining? *Mechanics built; larger sizes not yet run.* **Q4 — Faithfulness at scale.** Does swap-following keep rising with data and k , and can an auxiliary per-edge objective accelerate it? *Design sketched, not yet run.* **Q5 — Missing baselines.** An xRAG-style retriever-vector bridge trained under our exact protocol, per-entity prefix-tuning (the transductive upper baseline), and on-policy distillation (GKD [30]) as an objective ablation. *Not yet run.* **Q6 — Beyond 1-hop.** 2-hop neighborhoods raise the facts-per-token ratio ~ 10 – $100\times$ at fixed k ; we treat this as a rate-distortion ablation inside Q1, not a standalone contribution (the per-question subgraph-encoding literature is crowded [15, 17]). **Q7 — Second benchmark.** EntityQuestions integration for a second unseen-entity axis. *Loader stubbed.*

A compute-grant application (Swiss AI Initiative, CSCS Alps) is drafted to fund Q1–Q3; the pilot surface (Figure 2) is its preliminary-results centerpiece.

7 Reproducibility and infrastructure

Everything is public in the project repository: the data pipeline (snapshot, generation, tiering, teacher-path extraction; all resumable and cached), training (single 24 GB GPU per run), and the evaluation harness. Every metric in this report traces to a W&B run id (entity `university-of-zurich`, project `conceptformer-v2`) or a re-runnable CLI command, and every evaluation writes *per-item* outputs so any two systems can be compared with paired

statistics after the fact. The finding-by-finding evidence log (`docs/RESEARCH_FINDINGS.md`, F1–F16 plus methods notes M1–M7) records not only results but also the claims we *retracted* and why — start there, then `docs/RELATED_WORK.md` (annotated survey behind §2) and `docs/GRANT_PROPOSAL_DRAFT.md`. The three-command gate (`ruff`, `ty`, `pytest`; 244 tests) is green at every commit.

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